

Introduction

Variable stars are stars that undergo a significant variation in their apparent brightness over an observable length of time.

- Variable stars include stars that have an eruptive surface, have an occulting binary star, or are obscured by a halo of gas.
- Cepheid stars - supergiant stars with patterns of variability from 1-100 days
- T Tauri stars - young stellar objects with stochastic variations from accretion disks
- NSV stars - Potentially variable stars from past surveys, require more observation.



A star's variability in brightness over a time period

SKYNET is a global network of roughly two dozen telescopes, and consists of mostly optical telescopes.

- Their telescopes are made accessible to students through a credit system (1 credit = 1 second), where 1 credit is ~2 cents.

The documentation of variable stars is necessary for approximating celestial distances and understanding stellar evolution, including of our Solar System.

To resolve this research gap in NSVs, it was hypothesized that the potential variability of NSVs is determinable through the usage of SKYNET. The independent variable is the NSVs chosen for photometric analysis, and the dependent variable is the collected magnitudes across the experimental procedure.

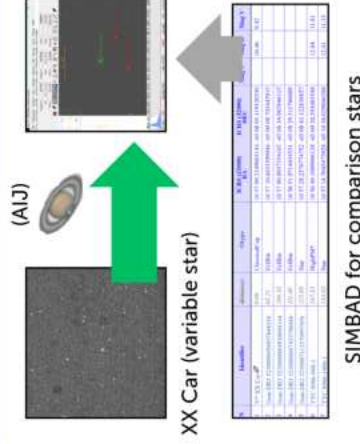
- A Cepheid and T Tauri star, to represent an ideal variable/irregular variable star, were observed as control variables.

The selected stars for the experimental procedure were XX Carinae, a Cepheid star, CT Cha, a T Tauri star, and two NSVs, NSV 3281 and NSV 2849.

Newly Identifying Stars: Determining the Periodic Variability of Classical Cepheids, T Tauris, and Suspected Variable Stars with SKYNET

Methodology

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|--|---|
| 1. Locate variable stars and irregular variable stars from the AAVSO variable star database. | Criteria - brighter than magnitude 13, with a variability period of <2 weeks. |
| 2. Commission exposures of selected stars through SKYNET daily. | 3. Use Astro ImageJ to identify magnitudes of my stars at certain timeframes. |
| 4. Create periodograms of data using phase-folding and power spectrums | 5. Download previous star data from AAVSO / SIMBAD, compare results. |



SIMBAD for comparison stars

Results

Data collected across 50 days of the experimental design process was compiled into a single Excel spreadsheet for periodogram analysis. Red boxes represent days where data was not collected.

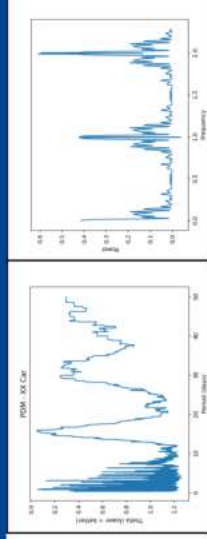
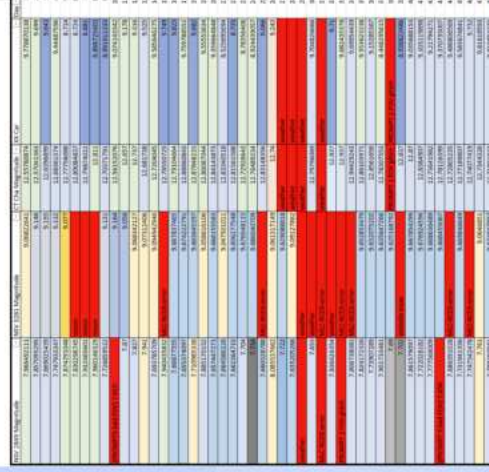


Figure 1: Phase Dispersion Minimization test on XX Carina

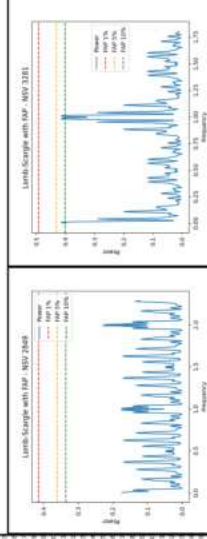


Figure 3: Lomb-Scargle + FAP test on NSV 2849



Discussion

Within Figure 1, the true period of XX Carina, 15.714 days, can be identified to be a peak in the PDM. Additionally, the Lomb-Scargle of CT Chamaeleon in Figure 2 produces a normal periodogram with minimal noise pollution. Both contal periodograms indicate the success of the photometric analysis on the selected stars.

A Lomb-Scargle was also conducted on NSV 2849, visible in Figure 3. A FAP test, indicating the validity of the peaks, was also included. For both the NSV 3281 and NSV 2849 periodograms, the peaks of suspected periodicity failed to surpass the 1% mark, which is the scientific norm for proving variability.

According to the data, both stars appear to be non-variable, shown by Lomb-Scargle.

Future Work

- Telescopes kept constant to mitigate noise issues for the observation of NSV 2849 and NSV 3281
- Revision of AIJ analysis methodology in a professional manner to reduce errors
- Revision of exposure lengths for each star to avoid oversaturation in the pictures taken

Works Cited

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